higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ROADS
NQF LEVEL 4

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This marking guideline consists of 6 pages.

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QUESTION 1

1.1
- To prevent stagnant or pools of water forming on top of a road surface.
- To prevent aquaplaning or other hazards, which could be a danger to motorists.
- Water can cause various types of damage to the road surface.
- Water can cause structural damage to the layers.
- Road drainage will prevent erosion on the sides of the road. (Any 3 × 1) (3)

1.2
- Outside the road shoulder breakpoint and away from the carriageway.
- Parallel to and away from the centre line of the road.
- Along a slope that is deeper than the road surface.
- Where it can direct the water away from the carriageway.
- Where it can direct water away from the road prism.
- Where it can direct water through a culvert away from the road prism.
- Where water runs along a toe of a fill to a point where it can be diverted away from the carriageway. (Any 5 × 1) (5)

1.3
- Stones (rocks) must be obtained from a reputable or commercial crushing source.
- Stones must be of correct quality and grading.
- Stones must be clean, hard, durable, dense and susceptible to rust.
- Correct quantity of stones for the completion of the gabion.
- Stones must not be too small as they will fall through the mesh holes.
- The stones should be tightly packed in the cage to make them more stable.
- The stones should be laid by hand.
- Stones' shapes and sizes must fit tightly together. (Any 4 × 1) (4)

1.4
- Water that softens the subgrade layers
- Erosion to underlying road layers
- Erosion to road shoulders and the road surface
- Contributes to cracking, potholes and deterioration of the road surface
- Capillary rise from natural water table
- Infiltration of water from adjacent irrigation, canals and dams
- High rainfall in low-lying areas (Any 5 × 1) (5)

1.5
- Large enough to cover the manhole
- Heavy enough to stay in place without being blown out of place by wind
- Strong enough to handle heavy loads
- Resistant to degradation over years of exposure to weather elements (3)

1.6
- Trenches (excavations) must be dug to depths of more than 300 mm below ground level.
- Trenches should be excavated to a minimal width.
- Excavations should be dug manually to ensure an accurate gradient.
- Trenches should be excavated to a consistent gradient.
- Excavations must be done when the soil is as dry as possible.
- Excavations should be provided with sufficient trench support to prevent (5)

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sides collapsing. (Any 5 × 1)

1.7 1.7.1 Any permeable textile material made from synthetic or natural fibres or materials. (2)

1.7.2 Separation√
It keeps layers of different soil particle sizes separate√ from one another. (Any ONE)

Reinforcement√
It strengthens the soil and increases the soil support around the geofabric material.√ (Any ONE)

Drainage
Allows water to pass either downward or laterally through the geofabric.√ (Any ONE)

Filtration√
Allows water to move through the geofabric while restricting soil particles passing through the fabric.√ (Any ONE) (3)
[30]

QUESTION 2

2.1
- Adequate friction
- Smoothness
- Noise control/low noise levels
- Rut resistance
- Shoving resistance
- Good drainage (Any 5 × 1) (5)

2.2
- Strength of the concrete to match the design requirements
- Uniform production of the concrete
- Durable to withstand in-service conditions
- Must be workable and be placed and compacted without segregation

(Any 3 × 1) (3)

2.3 2.3.1 **Workability**
The level of ease concrete can be worked with, manipulated and shaped (2)

2.3.2 **Consistency**
The ability of concrete to flow easily and to be worked with (2)

2.4
- Where heavy or concentrated wheel loads are to be carried
- Where a high frequency of turning and slewing movements is expected
- At high traffic volumes
- Where subgrade conditions are poor
- Where pavement must withstand severe-in-service conditions (temperature variations, oil or fuel spillage)
- Where access to below ground services is required (5)

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- Where appearance and aesthetic qualities are required (Any 5 × 1)

2.5
- Pen grade bitumen
- Cutback bitumen
- Modified bitumen
- Bitumen emulsion (4)

2.6
- Penetration test
- Viscosity test
- Flashpoint test
- Ductility test
- Solubility test
- Softening point test (Any 4 × 1) (4)

2.7 2.7.1 **Continuously graded asphalt**
- Low noise
- Very versatile and can be used for the structural components of the road and wearing course
- Good interlock of aggregate and well compacted
- Relatively low permeability if well compacted
- Strength and stiffness derived from binder and aggregate structure
- Can be used as thin overlays or thin surface course
- Cheaper than other bitumen mixture tyres (Any 3 × 1) (3)

2.7.2 **Gap-graded asphalt**
- Good aggregate interlock
- Strength and stiffness derived from binder and aggregate structure
- Best used in areas where rutting and cracking are of concern
- Relatively high binder contents provide good durability
- Low permeability (Any 3 × 1) (3)

2.7.3 **Open-graded asphalt**
- Good drainage capabilities and reduces water spray
- Reduces hydroplaning and provides good skid resistance
- High binder contents result in thick binder films
- Resistance to deformation and rotting
- Best applied in areas with fast traffic and minimal turns
- Absorbs and reduces tyre noise (Any 3 × 1) (3)

2.8 2.8.1 **Double seal**
- Tack coat
- First layer stone (aggregate)
- Penetration coat
- Second layer stone (aggregate) (Any 3 × 1) (3)

2.8.2 **Single seal**
A single layer of one-sized aggregate with a bitumen binder✓ sprayed over a road surface,✓ which has existing surface material.✓ (3)
[40]

QUESTION 3

3.1 3.1.1 **Road maintenance**
- To preserve or extend the service life of the road
- To slow down the deterioration of the road (2)

3.1.2 **Rigid pavement maintenance**
- Joint and crack sealing
- Slab stabilisation
- Diamond grinding
- Patches (Any 3 × 1) (3)

3.2 3.2.1 True
3.2.2 False
3.2.3 True
3.2.4 True
3.2.5 True
(Any 5 × 1) (5)

3.3
- Clearing and removing vegetation and debris from side drains
- Culvert cleaning
- Grass cutting
- Removing invader plants
- Fence repair work (Any 3 × 1) (3)

3.4 **Asphalt overlays**
- Clear the surface of all loose particles
- Use a water jet blaster to remove all surface markings
- Allow cleaned surface to dry
- Apply tack coat to the dried surface
- Spread and compact the overlay as specified
- Clear all loose aggregate after completion (Any 5 × 1) (5)

3.5 3.5.1 **Road reconstruction**
- Complete replacement of the entire road structure
- Apply new pavement layer (2)

3.5.2 **Road rehabilitation**
- Adding or replacing existing material on the existing road structure
- Restoration of original slope and natural drainage to prevent erosion
- Removal and restoration of original road carriageway
(Any 2 × 1) (2)

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3.6 3.6.1 **Traverse cracking**
Caused by indifferent slab lengths and deficient slab thickness,✓ moisture and temperature variations,✓ high thermal expansion✓ or insufficient base support.✓ (Any 2 × 1) (2)

3.6.2 **Shattered slabs and corner breaks**
Caused by insufficient slab thickness✓ and base support✓ (2)

3.6.3 **Punch-outs**
Caused by higher than allowable traffic loads,✓ insufficient structural strength,✓ deficient slab thickness✓ and insufficient subbase support.✓ (Any 2 × 1) (2)

3.6.4 **Deep spalling**
Caused by various forms of stress (rocks, water, etc.) in the joints,✓ cold weather,✓ bad curing,✓ heavy traffic✓ and weak joints.✓
(Any 2 × 1) (2)
[30]

TOTAL: 100